# Chapter 3 Notes

## 3.1

In the proof of Theorem 3.1, here is why the class of all transitive natural numbers is a set. We can show that it is a subset of $\omega$, and is hence a set by $A_2$. For suppose $x$ is a member of the class of all transitive natural numbers. Then obviously $x$ is also a natural number, hence $x \in \omega$, thus completing the proof.

Note that here is why the only natural numbers allowed for by Definition 2.2 are the ones defined right before Definition 1.1. By Definition 2.2, a natural number is a member of every inductive set. Now, consider the following inductive set: $\emptyset$ is an element, and if $x$ is an element, then so is $x^+$. Hence every natural number is a member of this set. But this set is just the set of all sets that were defined in the way just before Definition 1.1.

### Exercise 3.1

Suppose that $A$ is a class of natural numbers such that for every natural number $n$, if $A$ contains all numbers less than $n$, then $A$ contains $n$ as well. Now, we can prove by induction that $A$ contains all natural numbers. Consider the property $P$ meaning $A$ contains all numbers natural numbers less than or equal to $n$. Now, there are no natural numbers less than 0. Thus the statement that $A$ contains all natural numbers less than 0 is vacuously true; hence $A$ contains 0. Thus, because 0 is the only natural number less than or equal to 0, $P$ holds for 0. Now suppose that $P$ holds for some $n$, that is, $A$ contains all natural numbers less than or equal to $n$. But then $A$ contains all numbers less than $n+1$, meaning (by the definition of $A$) $A$ also contains $n+1$. Thus, by induction, $A$ contains all natural numbers.

### Exercise 3.2

Consider a natural number $n$.

(a): Suppose that $x \in n$. Now, it cannot be that $n = \emptyset$. Thus $n$ is non-empty, hence of the form $(n-1)^+$. Thus either $x \in n-1$ or $x = n-1$. Now, $\cup n$ is just the set of all elements of all elements of $n-1$ together with all elements of $n-1$. Now suppose $x = n-1$. Then $x \notin \cup n$. For suppose it were. Then it would either be an element of $n-1$ or it would be an element of an element of $n-1$. In the first case, we would have $n-1 \in n-1$, which is ruled out by Theorem 3.2. In the second case, there would be some $y$ such that $x \in y$ and $y \in n-1$. But then, by the transitivity of $n-1$ (by Theorem 3.1), $x \in n-1$, that is, $n-1 \in n-1$, which is again ruled out by Theorem 3.2. Thus $n-1 \notin \cup n$, meaning the statement is not necessarily true.

It should be noted that all elements other than $n-1$ of $n$ are members of $\cup n$. For all elements of $n$

are either elements of $n-1$ or equal to $n-1$. Thus, if they are not equal to $n-1$, the element is an element of $n-1$. But, because $n-1 \in n$, then all elements of $n-1$ are elements of $\cup n$.

(b): Suppose that $x \in \cup n$. Thus either $x \in n-1$ or $x$ is an element of some element of $n-1$. In the first case, because $n-1 \in n$, $x \in n$ (by the transitivity of $n$). In the second case, because $n-1$ is transitive, it follows that $x \in n-1$. But then, because $n-1 \in n$ and $n$ is transitive, $x \in n$. Thus, in either case, $x \in n$, and thus $\cup n \subseteq n$. Hence the statement is necessarily true.

(c): By part (b) of this problem, it follows that $x \in \cup n$ implies $x \in n$. But $n^+ = n \cup \{n\}$. Thus all elements of $n$ are in $n^+$, meaning $x \in n^+$. On the other hand, suppose that $x \in n^+$. Then it could be that $x = n$. But then, if $x \in \cup n$, we would have either $n$ being an element of an element of $n-1$, or $n \in n-1$. In the first case, the transitivity of $n-1$ would show that $n \in n-1$. Thus we would have, in either case, $n \in n-1$, which is impossible. For if $n \in n-1$, then, because $n-1 \in n$, $n-1 \in n-1$ by the transitivity of $n-1$, which is impossible (or, if $n-1 = \emptyset$, it could have no members). Hence it is not necessarily the case that $n^+ \subseteq \cup n$. Thus $\cup n = n^+$ does not necessarily hold.

Similar reasoning as to the above also shows that $n-1$ cannot be an element of $\cup n$. It should be noted that $n-1$ and $n$ are the only members of $n^+$ that are not members of $\cup n$. For any other element of $n^+$ is an element of $n$ but not equal to $n-1$. But $n = n-1 \cup \{n-1\}$. Hence it would be an element of $n-1$, and because $n-1$ is an element of $n$, it would also be an element of $\cup n$.

(d): On the one hand, suppose that $x \in \cup n^+$. Then either $x$ is an element of an element of $n$, or $x$ is an element of $n$. In the first case, because $n$ is transitive, $x \in n$. The second case means $x \in n$. Thus, in any case, $x \in n$. On the other hand, suppose that $x \in n$. Then $x \in \cup n^+$, as $n^+ = n \cup \{n\}$. Hence $\cup n^+ = n$.

(e): By part (b) of this problem, $\cup n \subseteq n$ holds. But by the answer to part (a) of this problem, it does not necessarily follow that $n \subseteq \cup n$. Thus it is not necessarily true that $\cup n = n$.

## Exercise 3.3

(a): On the one hand, suppose that $x \in \cup\omega$. Then $x$ is an element of a natural number. Thus either $x \in n$ or $x = n$. Now we prove the following statement: every element of every natural number is a natural number. The base case vacuously holds, as $\emptyset$ has no elements. Now suppose the statement holds for some $n$. Then $n^+ = n \cup \{n\}$. Hence, if $y \in n^+$, either $y \in n$ or $y = n$. In the former case, by the induction assumption, it follows that $y$ is a natural number, and in the latter case, $y$ would be a natural number (as $n$ is a natural number). Thus, in any case, $y$ is a natural number, therefore completing the proof. As a consequence of having proven this statement, it follows that $x$ is a natural number, and hence $x \in \omega$. On the other hand, suppose that $x \in \omega$. Thus, for some $n$, $x = n$. Now, all elements of $n^+$ are in $\cup\omega$. But $n$ is an element of $n^+$. Thus $n \in \cup\omega$, and hence $x \in \cup\omega$. Thus we have proven that $\cup\omega = \omega$.

(b): On the one hand, suppose that $x \in \cup\omega^+$. Now, $\omega^+ = \omega \cup \{\omega\}$. Thus either $x$ is an element of a natural number or $x$ is a natural number. By the results of part (a) of this problem, it follows that if $x$ is an element of a natural number, then $x$ is a natural number. Thus $x$ is a natural number, that is, $x \in \omega$. Now suppose that $x \in \omega$. Then $x \in \cup\omega^+$, as $x$ would be an element of $\omega$, which is an element of $\omega^+$. Thus we have proven that $\cup\omega^+ = \omega$.

Note that a generalization cannot be made of this problem. Define $\{\omega_0, \omega_1, \omega_2, ..., \omega_n, ...\}$ in the following way: $\omega_0 = \omega$ and $\omega_{n+1} = \omega_n^+$. Then it does not follow that $\cup\omega_n = \omega$ for all $n$. For consider $n = 2$. This is $\omega_1^+$, that is,

$$\omega_1 \cup \{\omega_1\} = (\omega \cup \{\omega\}) \cup \{(\omega \cup \{\omega\})\}$$

Thus $\omega \in \cup\omega_1^+$. But $\omega \notin \omega$, meaning $\cup\omega_2 \neq \omega$. Indeed, it is a general fact that, for $n \geq 2$, $\cup\omega_n \neq \omega$. To prove this, we can just prove that, for all $n \geq 2$, $\omega \in \omega_n$. We have just proven the base case. If the result

holds for some $n$, then $\omega \in \omega_{n+1}$, as $\omega_{n+1} = \omega_n \cup \{\omega_n\}$, and it was assumed that $\omega \in \omega_n$. Thus, for every $n \geq 2$, $\omega \in \omega_n$. Now we can prove the statement that, for $n \geq 2$, $\cup\omega_n \neq \omega$. We proved the base case for $n = 2$ above. Now consider some $n > 2$. Then $\omega_n = \omega_{n-1} \cup \{\omega_{n-1}\}$ and $\cup\omega_n$ consists of all elements of all elements of $\omega_{n-1}$ and all elements of $\omega_{n-1}$. Because $\omega$ is an element of $\omega_{n-1}$, it follows that $\omega \in \cup\omega_n$, and thus, because $\omega \notin \omega$, $\cup\omega_n \neq \omega$.

# Section 4

## Theorem 4.16

The proof of Theorem 4.16 seems incorrect to me. At a point in the proof, we rely upon Theorem 4.12. But that assumes $L$ is a bounded *subset* of $A$. However he assumes that $\emptyset \notin A$ and shows that $\emptyset \in L$, meaning $L$ is not a subset of $A$, contrary to the claim that it was.

## Exercise 4.1

We suppose that $N$ is closed under $g$, where $g$ is progressing, and that $b$ is an element of $N$ and that no proper sublcass of $N$ containing $b$ is closed under $g$. In order to prove the desired results, we can first prove (1) of Theorem 4.17. We are given that $N$ is closed under $g$. Call a sequence of the form $x, g(x), g(g(x)), g(g(g(x))), ...$ a $g - sequence$ (and call it a subset of a set $A$ if all elements of the sequence are elements of $A$). Label $x$ the first term of the $g$-sequence, $g(x)$ the second term, and, in general, if $y$ is the $n$th term of the $g$-sequence, then $g(y)$ is the $n + 1$th term of the $g$-sequence (and we say that $n$ is the *index* of the $n$th term of the sequence). We first show that $N$ is just the $g$-sequence $b, g(b), g(g(b)), ...$ Now, it is obvious that the $g$-sequence started by $b$ is a subset of $N$ (because $N$ is closed under $g$, and $b$ is in $N$). Then suppose that there was some element $c \in N$ outside of the $g$-sequence started by $b$. Then the $g$-sequence $c, g(x), g(g(c)), ...$ would be a subset of $A$. But then $b, g(b), g(g(b)), ...$ would be a proper subset of $N$ containing $b$ which was closed under $g$, contrary to the hypothesis. Hence the only elements in $N$ are the elements of the $g$-sequence started by $b$. Having shown this, recall that $g$ is a progressive function. Now consider any $x, y$ in $N$. If $x = y$, then $y \subseteq x$. Now suppose $x \neq y$. Let $x$ be the $m$th term of the $g$-sequence and $y$ the $n$th term. Thus $m \neq n$. On the one hand, suppose that $m < n$. An obvious induction argument shows that $x \subseteq y$. Now either $g(x) = y$ or not. In either case, $g(x) \subseteq y$ is obvious. On the other hand, suppose that $n < m$. Then the same argument shows that $y \subseteq x$, thus completing the proof of (1).

Here is the "obvious inductive argument" mentioned above (I will refer to this as Lemma 1). We prove that, for all $n$, all $m$th terms where $m < n$ are subsets of the $n$th term of the $g$-sequence. The base case is obvious (as there are no terms of index less than 1). Now suppose the result holds for the $n$th term $x$. Then all terms of index $m$ where $m < n$ are subsets of $x$. But then, because $g$ is progressing, $x$ is a subset of $g(x)$ (the $n + 1$th term), and because all the terms of index $m$ where $m < n$ are subsets of $x$, it follows that all terms of index $m$ where $m < n + 1$ are subsets of $g(x)$, completing the proof.

We now prove (2) of the problem. We prove by induction on $n$ that every non-empty subset of $N$ that is bounded by the $n$th term of $N$ (that is, the $n$th term of the $g$-sequence $b, g(b), g(g(b)), ...$) has a greatest element. The proof of this is similar to that of Theorem 4.12:

I was not entirely sure what (3) was trying to prove: I interpreted it to mean that, if $x$ is the greatest element of $N$, then it is a fixed point of $g$. Suppose that $x$ is the greatest element of $N$. Then, for all elements $y$ of $N$, $y \subseteq x$. Now, we know that $g(x) \in N$. Because $x$ is the greatest element of $N$, $g(x) \subseteq x$. But also, because $g$ is progressing, $x \subseteq g(x)$. Thus $g(x) = x$, meaning $x$ is a fixed point of $g$.

Now we prove (4) of the problem. Because we have shown that $g(x) \subseteq y$ or $y \subseteq x$ for all $x, y$ in $N$, it follows that $x \subseteq y$ or $y \subseteq x$ (as $x \subseteq g(x)$). Thus $N$ is a nest. It remains to be shown that every non-empty subclass of $N$ has a least element. So consider some non-empty subclass $A$ of $N$. Because $N$ is just the set of all terms of the $g$-sequence $b, g(b), g(g(b)), ...$, it follows that $A$ contains only members of that sequence. Hence $A$ contains some term $x$ whose index (in the $g$-sequence) $n$ is less than or equal to the indices of all the other elements of $A$. By Lemma 1, it follows that $x$ is a subset of all terms with an index greater than or equal to $n$, and hence $x$ is a subset of all other elements of $A$, meaning it is the least element of $A$.

Because $b$ starts the $g$-sequence $b, g(b), g(g(b)), ...$, which makes up the entirety of $N$, Lemma 1 above shows that $b$ is the least element of $N$ (is a subset of all elements of $N$). This proves (5) of the problem.

The reason that this generalizes some of the earlier results of the section is that it takes away the requirenment that $N$ be minimally inductive under $g$. For example, Theorem 4.7 holds now when $N$ is closed under $g$, where $g$ is progressing, and that $b$ is an element of $N$ and that no propert subclass of $N$ containing $b$ is closed. That is, we do not need to know that $N$ is minimally inductive under $g$.

## Exericse 4.2

(I did not think that this proof requires $A$ to be a set)
Suppose that $A$ is inductive under $g$ and that $A$ is a set. Thus we know
(1) $\emptyset \in A$,
(2) for every $x$ in $A$, the set $g(x)$ is in $A$.
Consider the $g$-sequence $\emptyset, g(\emptyset), g(g(\emptyset)), ....$ Because $A$ is inductive under $g$, that $g$-sequence is a subset of $A$. But clearly, that $g$-sequence is minimally inductive under $g$. Hence there is some subset of $A$ that is minimally inductive under $g$.